

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Architecture (New)

Semester: Fall

Year : 2024
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain about various hierarchical structure of computer system in detail. 7
- OR
- b) What is addressing modes? How operands of an instruction are stored using addressing modes?
- b) How data transfer micro operations takes place in registers? Explain direct and bus implementation of data transfer in register. 8
2. a) Explain about internal structure of processor. Also explain the design principle for modern system. 7
- b) Describe in detail about the hardwired control unit with block diagram. Why is it faster than micro-programmed control unit? 8
3. a) How floating point numbers are represented in computer and also explain about how addition and subtraction is performed for floating point number? 7
- b) Unsigned binary number cannot be multiplied directly in computer. Write the algorithm to multiply two unsigned binary numbers with example. 8
- OR
- Perform 2's complement binary division for -5/2 with its algorithm.
4. a) Prove that "Speedup factor for pipelined processor is equal to number of stages in pipeline." Assume that pipeline has $K = 8$ segment and execute $n = 212$ tasks in a sequence. Let the time taken to process the sub-operation in each segment is $35ns$. Calculate the speedup ratio. 8
- b) Draw the memory hierarchy with their relative characteristics and differentiate between main memory and auxiliary memory. 7
5. a) Why mapping is needed in cache? Explain advantages and disadvantages of each cache mapping techniques. 7

- b) Differentiate between GPU and TPU? Discuss Interrupt driven I/O with necessary block diagram. 8
- 6. a) Explain different interconnection structure in multiprocessor system with necessary block diagram. 7
- b) Compare and contrast Dual, Quad and Octa core computers. 8
- 7. Write short notes on: (Any two) 2×5
 - a) Future trends in computer
 - b) Flynn's Classification
 - c) I/O Processor and I/O Channels